

MUHAMMAD ARSHEYAM KHATTAK

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SUMMARY

BS Artificial Intelligence student at COMSATS University Islamabad with hands-on experience building full-stack web platforms, NLP tools, and data analysis utilities in Python and React. Completed coursework in data structures and algorithms, AI, and machine learning, and has independently built and shipped multiple ML and web projects, including an NLP pipeline for low-resource South Asian languages. Seeking an AI/ML internship to contribute technical skills and continue growing as a machine learning practitioner.

EDUCATION

BS Artificial Intelligence | COMSATS University Islamabad, Islamabad, Pakistan 2024 – Present
Relevant Coursework: Data Structures & Algorithms, Introduction to Artificial Intelligence, Machine Learning, Object Oriented Programming

PROJECTS

- NEXUS — Intelligent Urban Simulation Platform** | Python, Streamlit, Plotly, Search, Logic, CSP, Probabilistic Models, Planning
- Designed and built a real-time urban simulation and AI reasoning engine that integrates multiple AI paradigms — search algorithms, logic-based reasoning, constraint satisfaction problems (CSP), probabilistic models, and automated planning — into a single interactive city environment.
 - Modeled complex city-scale challenges such as traffic flow optimization, disaster response, resource allocation, and decision-making under uncertainty, demonstrating how different AI techniques can collaborate within one unified system rather than working in isolation.
 - Built an interactive Streamlit interface with Plotly visualizations, allowing users to observe simulated city dynamics, AI-driven decisions, and system responses to changing conditions in real time.
- Crypto Careers** | React, Python, CoinGecko API, News API
- Built a full-stack cryptocurrency market analysis platform that pulls live data from the CoinGecko API to display real-time prices, daily percentage movements, and trending coin rankings updated continuously as markets change.
 - Integrated the News API to curate and surface relevant crypto news and articles alongside market data, giving users both quantitative price context and qualitative market sentiment in one place.
 - Developed a responsive, component-based front end in React for a smooth browsing experience across devices, paired with a Python backend responsible for fetching, cleaning, and serving data through internal endpoints.
- CSV Analyzer** | Python, Pandas, NumPy, Matplotlib
- Built an interactive data analysis utility that lets a user upload any CSV file and instantly receive a full statistical profile of the dataset, including descriptive statistics (mean, median, standard deviation) and automatic missing-value detection.
 - Used Pandas and NumPy to process and clean uploaded data on the fly, and Matplotlib to auto-generate distribution histograms and correlation heatmaps, helping users spot patterns and relationships in their data without writing any analysis code themselves.
- Low-Resource Language NLP — Urdu/Pashto Text Processing** | Python, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLTK
- Built a machine learning pipeline specifically for low-resource South Asian languages, tackling the core challenge of data scarcity through custom corpus collection, text cleaning, and tokenization strategies adapted for the non-Latin Nastaliq script used by Urdu and Pashto.
 - Developed classification and text-normalization models using scikit-learn, then used Matplotlib/pyplot and Seaborn to visualize class distributions, token-frequency patterns, and model performance across different dialect variants — surfacing where the models struggled most with underrepresented dialects.

CERTIFICATIONS

Supervised Machine Learning: Regression & Classification DeepLearning.AI / Stanford Online	Aug 2025
Machine Learning and NLP Basics Edureka	Aug 2025
Adobe Premiere Pro for Beginners: Quickstart Video-Editing Coursera Project Network	Aug 2025
AI For Everyone DeepLearning.AI	Aug 2025

ACTIVITIES & LEADERSHIP

Assistant Director of Graphics Team — IEEE Society | COMSATS University Islamabad

- Lead the graphics team, overseeing visual design and creative direction for society events and campaigns.

SKILLS

Languages: Python, Java, C++, C, HTML, CSS
ML & Data: scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLTK
Tools & Frameworks: React, Qt, CMake, Tkinter, VS Code, Git/GitHub, Cursor, Claude Code
Design: Adobe Photoshop, Adobe Illustrator, Adobe Premiere Pro, Figma, Canva
Other: Data Structures & Algorithms, AI Search Algorithms, GUI Development, Graphic Design